

Name:

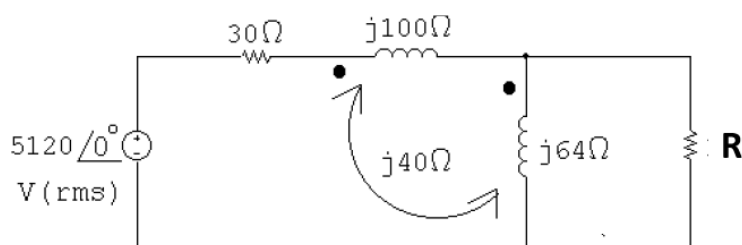
Class serial No.

Answer all questions. Write the answer in the space provided.

1. Find the average power in a resistance $R = 10 \Omega$ if the current is $i = 10\sin\omega t + 5\sin 3\omega t + 2\sin 5\omega t$ A

2. Two identical coils have an equivalent inductance of 0.08 H in series aiding and 0.035 H in series opposing. Find the coupling coefficient.

3. Find the average power dissipated in the resistance R . $R = 160 \Omega$



4. An unbalanced set of voltages are given as follows:

$$V_a = 120\angle 15^\circ, V_b = 30\angle -90^\circ, \text{ and } V_c = A\angle 165^\circ$$

Find the negative sequence voltages if $A = 90$

5. What are the performance requirements of control system? Give one specification of stability in time domain and frequency domain.

6. For an open loop system, if the magnitude plot crosses 0 dB line at 3.55 rad/sec and phase plot crosses -180 line at 3.32 rad/sec. If the gain at 3.32 rad/sec is 1.34 dB and phase at 3.55 rad/sec is -184.74 degree determine the phase margin. What is the actual gain when the root locus of this system crosses the imaginary axis.

7. What are the four broad topics in the Control System course which you will study in the next semester?